

계산뇌과학입문(이) (LS411-01)

학기	2025년 1학기
담당교수	유우경
이수구분	교선평수
학점	3.0
강의시간	수10:30-12:00(E4 - 113), 금10:30-12:00(E4 - 113)
강의실	E4 514 (교수실)
소속	기초학부

교과개요

This course covers essential concepts and methodologies for understanding how the brain processes information and generates behavior. Key topics include neural modeling, neural network architectures, synaptic plasticity, neuronal dynamics, learning and memory, and the application of computational techniques to study brain function. The course provides a foundation to understand and engage with the principles and computational methods used to simulate and analyze brain activity.

교과목표

You will learn the introduction and motivation of simplified models of neurons that are suitable for exploring information processing in large brain-like networks. Additionally, it introduces several fundamental network architectures and discusses their relevance for information processing in the brain, giving some examples of models of higher-order cognitive functions to demonstrate the advanced insight that can be gained with such studies.

주차별 강의계획

- 1주차: Class Introduction
- 2주차: preparation of simulation exercise
- 3주차: A first simple neuron model
- 4주차: The Hodgkin-Huxley model and detailed ion-current based neuron models
- 5주차: Synapses, dendrites and the cable equation
- 6주차: Two-dimensional models and phase plane analysis
- 7주차: Modern phenomenological neuron models
- 8주차: Mid-term exam
- 9주차: Neuronal Populations
- 10주차: Fokker-Planck equation for stochastic integrate-and-fire neurons
- 11주차: Associative Memory in a Network of Neurons
- 12주차: Attractor Networks and Generalizations of the Hopfield model
- 13주차: Continuum models: Cortical Fields and Perception
- 14주차: Decision models: Competitive Dynamics
- 15주차: Synaptic plasticity and learning
- 16주차: Final exam

학습윤리

Dishonest academic practices, such as cheating and plagiarism, are unacceptable. Engaging in such behaviors will lead to receiving no points on the exams and will be addressed as per the university's disciplinary standards.

교과정책(방침)

Cell phones should be turned off or set to silent mode during class. No texting or browsing on phones during class.